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Student Conference Paper

Multi-Class Skin Lesion Classification Using Deep Convolutional Neural Networks

A Transfer Learning Approach on ISIC 2019 Dataset

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Keywords: Skin lesion classification, melanoma detection, dermoscopy, convolutional neural networks, transfer learning, ISIC dataset

Abstract

Skin cancer represents a significant global health burden, with melanoma being particularly aggressive due to its metastatic potential. Early diagnosis dramatically improves survival rates—approximately 99% when detected early versus only 20% in advanced stages—making accurate classification systems essential.

This study presents a deep learning framework for automated multi-class skin lesion classification using dermoscopic images. We utilize the ISIC 2019 Challenge Dataset containing 25,331 images across eight diagnostic categories. The dataset exhibits severe class imbalance, with benign nevi constituting 50.8% of samples while rare lesions such as dermatofibroma represent only 0.9%.

To address these challenges, we employ transfer learning with ResNet-50 and EfficientNet-B3 architectures pretrained on ImageNet, combined with focal loss optimization and data augmentation techniques. Our approach demonstrates promising results, achieving [XX]% overall accuracy and [0.XX] AUC-ROC for melanoma detection.

The developed system shows potential as a clinical decision support tool, enabling efficient preliminary screening and reducing diagnostic variability in dermatological practice.

Keywords: Skin lesion classification, melanoma detection, dermoscopy, convolutional neural networks, transfer learning, ISIC dataset, class imbalance, deep learning.

1 Introduction

General Introduction to the Topic

Skin cancer is one of the most prevalent malignancies worldwide, with incidence rates continuing to rise across diverse populations. Among various types, melanoma stands out as particularly dangerous due to its tendency to metastasize to distant organs. The prognosis differs dramatically based on detection timing: early-stage melanoma has a five-year survival rate of approximately 99%, whereas late-stage disease with metastasis drops to around 20%.

Currently, dermatologists primarily diagnose skin cancer through visual examination of pigmented lesions. However, this approach is inherently subjective and prone to considerable inter-observer variability, even among experienced clinicians. These limitations have motivated increasing research into automated diagnostic systems that can provide more objective and consistent assessments.

Literature Review

Deep learning methods have gained substantial attention for skin lesion analysis in recent years. In a landmark study published in Nature, Esteva et al. [2] demonstrated that a single CNN trained end-to-end on dermoscopic images could achieve dermatologist-level classification accuracy across 757 classes of skin conditions. Their model performed comparably to 21 board-certified dermatologists in binary classification tasks.

Subsequently, Haenssle et al. [3] conducted a prospective reader study comparing a CNN against 58 dermatologists. The deep learning system achieved superior specificity (82.5%) compared to both less experienced (71.3%, $p < 0.01$) and more experienced (75.7%, $p < 0.01$) dermatologists. In a comprehensive systematic review, Brinker et al. [4] analyzed CNN-based skin lesion classification research, finding consistent patterns of

superior or equivalent performance compared to human experts. These studies collectively establish the feasibility of deep learning for automated skin lesion diagnosis.

Research Problem and Gap

Despite promising advances, several critical challenges hinder widespread clinical adoption of these technologies. The most significant obstacle is the pronounced class imbalance inherent to dermatological datasets. In the ISIC 2019 dataset, benign nevi (NV) comprise over half of all images (50.8%), while rare lesions like dermatofibroma (DF) account for less than 1% (0.9%).

This imbalance creates substantial difficulties for standard deep learning optimization strategies, which typically assume approximately balanced training data. Additionally, skin lesions exhibit considerable morphological heterogeneity due to variations in imaging protocols, illumination conditions, and patient demographics. These factors necessitate specialized architectural designs and training methodologies tailored specifically to dermatological imaging characteristics.

Proposed System Overview

This study presents a comprehensive deep learning framework designed to address these challenges in multi-class skin lesion classification. Our system integrates transfer learning with class-imbalance-aware training strategies to classify lesions into eight categories: melanoma (MEL), melanocytic nevus (NV), basal cell carcinoma (BCC), actinic keratosis/intraepithelial carcinoma (AKIEC), benign keratosis (BKL), dermatofibroma (DF), vascular lesions (VASC), and squamous cell carcinoma (SCC).

We selected two established CNN architectures: ResNet-50 [7] and EfficientNet-B3 [8], both pretrained on ImageNet. To address class imbalance, we employ focal loss [9], which reduces the contribution of easy-to-classify examples while focusing training on hard, misclassified samples. The framework also incorporates strategic data augmentation to enhance model generalization without requiring additional data collection.

Technical Infrastructure

ResNet-50, introduced by He et al. [7], utilizes residual learning with skip connections that facilitate gradient flow through deep networks. This architecture has proven particularly effective for medical image analysis due to its hierarchical feature representation capabilities, capturing both fine texture details and larger morphological structures.

EfficientNet-B3, developed by Tan and Le [8], employs compound scaling that uniformly adjusts network depth, width, and resolution. This approach achieves high accuracy with significantly fewer parameters—for example, EfficientNet-B7 attains 84.3% top-1 ImageNet accuracy while being 8.4× smaller and 6.1× faster than previous state-of-the-art models. Both architectures benefit from ImageNet pretraining, enabling transfer of generic visual features subsequently fine-tuned for dermatoscopic images.

Example Applications

Deep learning has demonstrated clinical utility across multiple medical imaging domains. In dermatology, automated melanoma detection systems serve as decision support tools, providing quantitative risk assessments that complement dermatologist evaluations. These systems also enable teledermatology applications, allowing preliminary screening of lesions in remote or underserved regions.

A notable parallel development is CheXNet [10], developed by Rajpurkar et al. for pneumonia detection in chest X-rays using a 121-layer DenseNet architecture. Recent investigations have explored multimodal approaches combining dermoscopic analysis with clinical photographs and patient metadata, while ensemble methods combining multiple deep learning models with human experts show promise for collaborative intelligence

systems.

Comparison with Previous Work

Traditional skin lesion classification relied on handcrafted features following the ABCD criteria (asymmetry, border irregularity, color variation, diameter). While these methods provided interpretable rules aligned with clinical practice, they suffered from limited generalizability across diverse lesion types and imaging conditions.

Modern deep learning approaches autonomously learn hierarchical feature representations directly from raw pixel data, discovering patterns that may not be explicitly described in clinical literature. This capability enables classification of complex lesions that defy simple morphological characterization. Comparative studies consistently demonstrate that end-to-end trained CNNs substantially outperform traditional pipelines using handcrafted features, particularly for differentiating visually similar lesion types.

Original Contribution

This research makes three primary contributions to automated skin lesion classification. First, we present a detailed comparison of ResNet-50 and EfficientNet-B3 architectures specifically for multi-class classification of the eight ISIC 2019 categories, offering practical guidance for architecture selection.

Second, we implement and evaluate focal loss optimization combined with class-weighted sampling to explicitly address class imbalance, demonstrating measurable improvements in detecting underrepresented malignant lesions. Third, we conduct thorough evaluation using clinically relevant metrics including sensitivity, specificity, and AUC-ROC, providing detailed insights into model performance across diverse lesion types.

Itemized Contributions

The specific contributions of this paper are:

- Development of a transfer learning framework using ResNet-50 and EfficientNet-B3 for eight-class skin lesion classification on the ISIC 2019 dataset.
- Implementation of focal loss optimization with class-weighted sampling to address severe class imbalance (NV: 50.8% versus DF: 0.9%).
- Comparative analysis of architectural performance and computational efficiency to inform deployment decisions.
- Evaluation using clinically relevant metrics (sensitivity, specificity, F1-score, AUC-ROC) across all eight categories.
- Investigation of data augmentation strategies to enhance model robustness while preserving clinical features.

Paper Structure Overview

The remainder of this paper is organized as follows. Section 2 reviews related work in skin lesion classification, emphasizing deep learning methods and class imbalance handling. Section 3 provides background on the architectures and optimization techniques employed.

Section 4 describes the dataset characteristics, preprocessing, experimental setup, and evaluation protocols. Section 5 presents experimental results with comparative analyses. Section 6 discusses findings, limitations, and future work. Finally, Section 7 concludes with a summary of key findings and their clinical significance.

2 Dataset Description

2.1 ISIC 2019 Challenge Dataset

This study utilizes the ISIC 2019 Challenge Dataset [1], publicly available from the International Skin Imaging Collaboration (ISIC). This dataset was compiled for the 2019 skin lesion classification challenge and represents one of the largest publicly available collections of dermoscopic images.

The dataset contains 25,331 images distributed across eight diagnostic categories. It aggregates data from multiple established repositories, primarily the HAM10000 dataset [5] (10,015 images) and the BCN20000 dataset [6] (19,424 images). HAM10000 includes multi-source dermatoscopic images from various clinical centers, while BCN20000 contains images captured between 2010–2016 at the Hospital Clinic de Barcelona under "in the wild" conditions. This aggregation incorporates variations in imaging equipment, protocols, and patient populations, enhancing the dataset's representativeness of real-world clinical scenarios.

2.2 Diagnostic Categories and Class Distribution

The dataset encompasses eight diagnostic categories defined by the ISIC 2019 challenge organizers:

1. **Melanoma (MEL):** Malignant tumor of melanocytes; the most aggressive form of skin cancer with high metastatic potential.
2. **Melanocytic Nevus (NV):** Benign proliferation of melanocytes, commonly known as moles. This category represents the majority class (50.8%).
3. **Basal Cell Carcinoma (BCC):** The most common form of skin cancer, characterized by slow growth and local invasiveness.
4. **Actinic Keratosis and Intraepithelial Carcinoma (AKIEC):** Precancerous lesions and early-stage squamous cell carcinoma in situ.
5. **Benign Keratosis (BKL):** Non-cancerous skin growths including seborrheic keratosis and solar lentigo.
6. **Dermatofibroma (DF):** Benign fibrous skin lesions, typically appearing as firm nodules on extremities.
7. **Vascular Lesions (VASC):** Benign vascular anomalies including hemangiomas.
8. **Squamous Cell Carcinoma (SCC):** Malignant tumor arising from keratinocytes; the second most common skin cancer.

A significant challenge of this dataset is severe class imbalance. The NV class dominates with 12,875 images (50.8%), while DF contains only 239 images (0.9%). This skewness presents substantial challenges for model training, as standard optimization procedures tend to bias toward majority classes while underperforming on rare but clinically critical categories such as melanoma.

2.3 Data Characteristics and Preprocessing

All images are dermoscopic photographs captured using specialized equipment that illuminates and magnifies skin lesions, enabling visualization of subsurface structures. The images vary in resolution, lighting, and anatomical location, reflecting the diversity of real clinical acquisition scenarios.

For this study, the dataset was partitioned using stratified sampling to maintain class distribution proportions across subsets. We allocated 80% for training, 10% for validation, and 10% for testing. This stratified approach ensures each subset contains representative samples from all eight categories, facilitating reliable model evaluation even on underrepresented classes.

2.4 Ethical Considerations and Data Availability

The ISIC 2019 Challenge Dataset is publicly available at `https://challenge.isic-archive.com/data/` [1] and distributed under terms appropriate for academic research. The dataset has been de-identified to protect patient privacy in accordance with ethical guidelines for medical image sharing. Research use has been authorized through the ethical review boards of participating institutions, as documented in the original dataset publications [5, 6].